Rupesh Sen

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Summary

Engineer working on AI products, from testing models to building features and solving problems with clients. On an enterprise voice AI platform, I evaluate LLM and speech-to-text models, optimise prompts (38% fewer tokens, 20% lower latency), and work directly with clients to find the root cause of issues and get them fixed with engineering. I build with Python, SQL, React, Node.js and Django, and have trained deep learning, computer vision and NLP models (minor in Computational Intelligence). Comfortable moving between code, client conversations and product decisions.

Experience

Product Intern (AI) **Sep 2026 – Present**
Blue Machines AI (Apna.co) – Enterprise multilingual voice AI platform *Bengaluru, India*

- Improved **usability across the client-facing platform**: reviewed user journeys end to end, identified pain points in key flows, and worked with delivery and engineering teams to make features more intuitive for enterprise clients.
- Organised **role-based access control (RBAC)**: grouped existing platform features into role-based sets with no hidden dependencies between them, and regrouped related settings so each client role sees what it needs.
- **Evaluated LLMs** on response quality, context handling and token efficiency, and **speech-to-text models** on complex-word accuracy and speech chunking, weighing quality, cost and latency trade-offs; rewrote prompts to **cut LLM token usage by 38%** and **response latency by 20%** (measured before and after).
- Analysed **call funnels and drop-off points** to find where callers drop off, and shared findings to improve agent conversation flows.
- Own daily **bug triage and root cause analysis** for client-reported issues, raising detailed tickets to engineering; run agent testing and write client-facing documentation.

Software Engineer Intern **Jan 2026 – Mar 2026**
nanoFarm – Agri-tech

- Rebuilt the company's web platform (React, Tailwind CSS, Node.js) with a focus on page performance and user experience.
- Shipped **AI features for farmers**: an OpenAI-powered query assistant and a plant-disease detection CNN model served via REST API.

Software Development & Operations Intern **Jun 2024 – Aug 2024**
Nabin RCB *Birbhum, West Bengal*

- Digitised manual accounting and inventory workflows with a Django/PostgreSQL system, **cutting data entry time by 60%** and **improving inventory turnover by 25%**.
- Automated sales data validation with SQL and Python, making sales and data-entry work **15% more efficient**.

Projects

TiffinTrack – Two-sided meal subscription marketplace | *React, Node.js, PostgreSQL, WebSockets, Groq LLM* GitHub

- Designed separate apps for two user groups, customers and meal providers, around each side's needs: real-time order updates, location-based discovery and subscription management.
- Added an **AI chat assistant (LLM + RAG)** for customer queries, review sentiment analysis, and analytics dashboards to help providers track and improve performance.

CodeFolio – Unified coding-profile dashboard | *React, TypeScript, OAuth, REST APIs* Live | GitHub

- Solved scattered coding profiles by combining LeetCode, GeeksforGeeks and GitHub progress into one analytics dashboard, with OAuth sign-in and CI/CD on Vercel.

Skills

Product: Root cause analysis (RCA), problem discovery & user journey mapping, feature prioritisation, requirements definition, success metrics & KPIs, funnel & conversion analysis, A/B testing, feature testing, cross-functional collaboration
AI & LLMs: LLM & STT evaluation, cost/latency trade-offs, prompt engineering, RAG, generative AI, voice agents
ML & Deep Learning: PyTorch, TensorFlow, CNNs, computer vision (YOLO, OpenPose), NLP & sentiment analysis, predictive modelling, classification models, data preprocessing
Data & Tech: SQL, Python, Power BI, dashboards, REST APIs, React, Node.js, Django, PostgreSQL, GCP, Git

Education

Manipal Institute of Technology **2022 – 2026**
B.Tech in Computer Science & Engineering (Minor: Computational Intelligence) *Manipal, Karnataka*

- Relevant coursework: Software Engineering, Business Analysis, Machine Learning, Database Management Systems

Certifications & Leadership

- IBM Generative AI Fundamentals Professional Certificate (Coursera, 2025) | Business Analysis & Process Management (Coursera, 2025)
- Executive & Associate, E-Cell Manipal (Sep 2022 – May 2024): coordinated 5+ technical workshops and managed stakeholder engagement for 500+ participants.